A COMPARATIVE FIBER AND NUMERICAL ANALYSIS OF THE PYRAMIDAL TRACT

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ONE FIGURE

A considerable number of degeneration and physiologic experiments have been done on the pyramidal tract of various mammals without taking into account the nature of its inherent fibers. The only quantitative methods employed on this part of the nervous system have been either general inspection of the pyramids or measurements of areas, both of which give an incomplete picture. It was felt that the application of a refined silver stain to the lower part of the medulla in a series of mammals would give comparative information on the size, number, and speed of conduction of the pyramidal tract fibers. Some clue might be gained also about the disputed cells of origin of this projection system because it is believed the cellular sizes should be proportional to the caliber of the component fibers.

MATERIAL AND METHODS

Eight animals were selected for the study. They include the opossum, mouse, rat, rabbit, cow, cat, dog, and man. The values obtained by Lassek and Rasmussen ('39) for man are used to complete the series. Two medullas were utilized for each species to minimize the effects of individual variations with the exceptions of the opossum and cow which involves only one and man where three specimens were studied.

The only constant characteristic of the pyramidal tract in the series is that it decussates in the lower part of the medulla.

For this reason, it was necessary to select a level for study just rostral to the crossing. All results and conclusions given below are based on the findings at this one plane. The pyramidal bundle is situated in the lateral funiculus in the cat, dog, and man. In the mouse and rat, it occupies the posterior funiculus. It extends for only a short distance into the cervical cord in the opossum and rabbit.

The silver stain selected is the one recently developed by Davenport and his co-workers ('38, '39). It has been the technic of choice in our hands and appears to have a selective affinity for both the finer and coarser fibers of the pyramidal tract. The area of each pyramid was measured by means of an Edinger-projection apparatus. Shrinkage was carefully determined for each specimen so the values given for the areas approximate normal size. The stained axis-cylinders were examined as to their diameter with an ocular micrometer. The width of obvious medullated fibers could be calculated by following the rule that the axis-cylinder is equal to one-third the total thickness. For ascertaining the number of fibers present in the tract, a Whipple square was used. All of the fibers were counted in the mouse; one-quarter in the rat, rabbit, cat, dog, and opossum; and one-eighth in one and one-sixteenth in two humans. All sections were inspected and counted at a magnification of 1,125 times.

RESULTS AND DISCUSSION

The results and discussion are considered under three headings: numerical values, sizes, and cells of origin of the pyramidal tract.

Numerical values of the pyramidal tract

The approximate number of fibers computed to be present in the pyramidal tract of each of the animals at a level just rostral to the motor decussation is as follows: man, 1,101,000; dog, 285,300; cat, 186,000; rabbit, 101,700; opossum, 75,700; rat, 73,000; and mouse, 32,300 (table 1). We did not count the fibers of the cow's pyramidal tract because of the difficulty in

outlining exactly its boundaries but we estimate it probably contains around a half million fibers.

In the three humans studied, the square area is much the largest in the series, averaging 11.4 square millimeters. Each human pyramid gave numerical values exceeding a million. This augments by about four times the count obtained by Weil and Lassek ('29) with the Bielschowsky technic. The increase is believed to be due to a greater specificity of the stain used in the present study for the smaller fibers and secondly to the fact that we are counting at a higher level.

TABLE 1

The square area, approximate number of fibers within the pyramidal bundle, the calculated number of pyramidal tract fibers per square millimeter, and the unit ratio of pyramidal tract fibers using man as a standard. All numerical values were determined on the basis of a silver study of the pyramidal tract just rostral to the motor decussation

ANIMAL	AREA OF PYRAMIDAL TRACT IN SQUARE MILLIMETERS	APPROXIMATE NUMBER OF FIBERS IN PYRAMIDAL TRACT	CALCULATED NUMBER OF PYRAMIDAL TRACT FIBERS PER SQUARE MILLIMETER	RATIO TO MAN
Opossum	0.28	75,700	272,400	2.8:1
Mouse	0.067	32,300	480,600	5.0:1
Rat	0.30	73,000	243,100	2.5:1
Rabbit	0.28	101,700	363,600	3.8:1
Cat	1.13	186,000	164,600	1.7:1
Dog	3.49	285,300	81,900	0.8:1
Man	11.43	1,101,000	96,500	1.0:1

More than this number, 1,200,000 myelinated fibers, have been reported by Arey and Bickel ('35) in the optic nerve of man which has about the same square area as the pyramidal tract at the level we have chosen.

The numerical value given for the dog, 285,300 fibers in an area of 3.5 sq. mm., is the average of two counts on animals of widely different weights. One scaled 20 and the second 70 pounds. The count of the larger dog exceeded that of the smaller by 100,000. It may be that the size of the animal in this species is of some numerical significance. It is also possible that the number of pyramidal tract

fibers may be proportional to the mass in any animal where there is a marked variation in this respect. Another factor to be considered is that in the smaller dog the fibers of the pyramidal tract were possibly not as efficiently stained. This question is being pursued further. In the dog, the area is 30.5% and the number of pyramidal tract fibers 25.9% that of man. The motor cortex of the dog is well developed with hindleg, foreleg, neck, and facial muscles represented therein. In contrast, digital movements are not highly specialized. It is interesting to speculate on the necessity for such a large number of pyramidal tract fibers in an animal with unspecialized digital movements such as is the case in the dog. From the locomotor viewpoint, it is a highly active animal which does a considerable amount of running.

Considering the size of the cat, the pyramidal tract fibers are numerous, 186,000 being present in 1.1 sq. mm. of area. The area is 9.9% and the number of fibers 16.9% that of man. The cat has a well developed motor cortex which is situated in the anterior and posterior gyri. McKibben and Wheelis ('32) found that the gyrus proreus also contributes fibers to the pyramidal tract of this animal.

Stimulation experiments on the motor cortex of rodents have given results which indicate it is not well defined. Investigators are not in agreement as to whether there is a cortical region for the hindleg muscles. Swank ('36) in his recent work on the pyramidal tract of the rabbit could find no hindleg area and he traced the fibers arising from the motor cortex only as far as the first cervical segment of the cord. He also described fibers present in the pyramidal tract which arose from the caudal two-thirds of the lenticular nucleus and extended as far distally as the cervical enlargement. It is possible, therefore, that the 100,000 fibers present in 0.28 sq. mm. of pyramidal tract just above the motor decussation in the rabbit represent fibers from at least two sources, the motor cortex and lentiform nucleus. These fibers, too, must terminate suddenly over a short distance in the cervical segments. Here the square area of the pyramidal

tract is 2.4% and the number 9.2% that of man. In the rat and mouse, the pyramidal tract can be traced easily into the lumbosacral regions of the spinal cord. The axis-cylinders are relatively numerous in both, the rat having 73,000 in 0.3 and the mouse 32,300 in 0.067 sq. mm. of pyramidal tract. The square area of the rat is 2.6% and the numerical value 6.6% that of man. In the mouse, the equivalent values are 0.6% and 2.9 % respectively.

The pyramidal tract of the opossum exhibits some similarity to the rabbit. No hindleg area seems to be present in the motor cortex and the pyramidal tract extends only to the level of the fourth cervical segment (Turner, '24). Here again, the 75,700 axis-cylinders contained within 0.28 sq. mm. of pyramidal tract area which arise from the motor cortex must terminate abruptly after they decussate. The square area of the pyramidal tract of the opossum is the same as the rabbit and forms 2.4% while the number calculated to be present is 6.9% that of man.

The sizes of the pyramidal tract fibers can now be discussed as they are an important factor in the high counts reported above.

Size of pyramidal tract fibers

Although the pyramidal tract is variable in its size, length, and position, all those examined in the series have one characteristic in common. That is, that the predominant axis-cylinder is one of small caliber (fig. 1).

Some general idea of the magnitude and condensation of pyramidal tract axis-cylinders can be gained by computing the number of fibers per unit area. The standard of measurement selected for this calculation was 1 sq. mm. In this connection, the mouse would have 480,600 fibers per square millimeter, the rabbit 363,600, the opossum 272,400, the rat 243,100, the cat 164,600, man 96,500, and dog 81,900 (table 1). Another way to express this would be to say that the mouse has 5.0 times more pyramidal tract fibers for the same given area than does man. The rabbit would have 3.8, the opossum 2.8, the rat

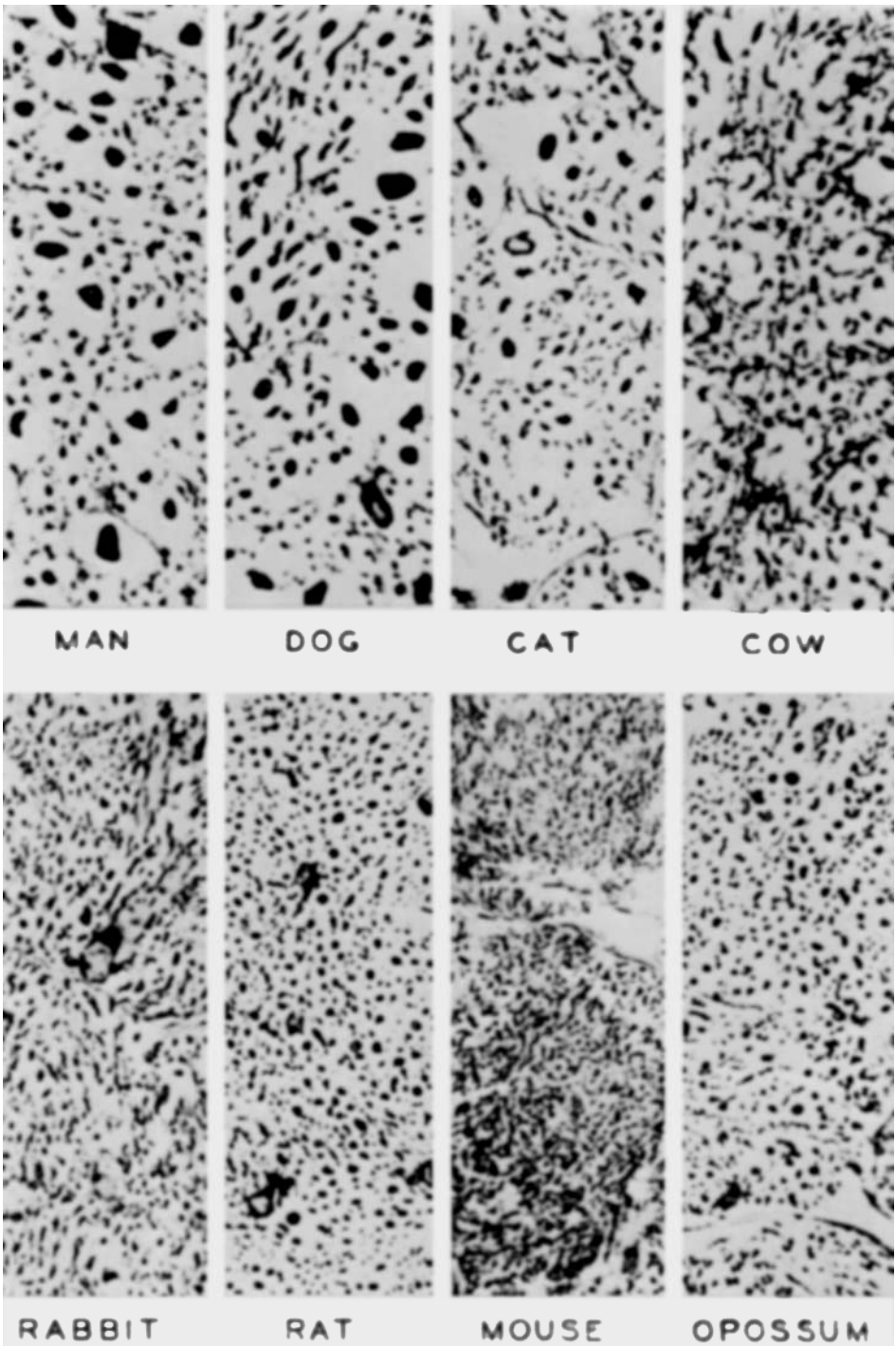
2.5, the cat 1.7, and the dog 0.8. Because of these variations in the diameter of the axis-cylinders, a knowledge of the areas does not give a true conception of the status of the tract.

Man possesses the largest axis-cylinders in the series (fig. 1). These large fibers are roughly between 10 and 25 μ and are few in number. They form about 4% of the total, or 40,000, which value corresponds closely with the 25,000 Betz cells counted by Campbell ('05). About 96% of the fibers are smaller than 10 μ in diameter and by far the greater majority of these will be between 1 and 5 μ.

The silver picture of the pyramidal tract of the dog is very similar to that of man (fig. 1). It seems to have more intermediate sized fibers which together with more separation accounts for the fact that there is a lesser number of fibers per square millimeter. Because at first there was some indication that the bigger mammals possessed larger fibers, we obtained and examined the pyramidal tract of the cow. We were curious, too, to know what the silver stain would show on this bundle in a hoofed animal which has no digital movement. Although the cow weighs ten to twenty times more than man, the axis-cylinders are extremely small and it simulates the picture of the pyramidal tract of the smaller animals (fig. 1) except that the myelinated fibers appear to be more numerous and there is consequently more spacing of the axis-cylinders. The pyramidal tract of this ungulate would have to subserve a different function than that of digital movement. The cat differs from man and dog in possessing no large sized axis-cylinders. There are a few fibers of medium caliber but most are of small diameter.

In the remaining smaller animals of the series (rat, opossum, rabbit, and mouse), the pyramidal tract presents much the same picture. The axis-cylinders are all small and closely packed together. The mouse has the smallest fibers of the

Fig. 1. Untouched photomicrographs of silver-stained pyramidal tracts in a series of eight mammals all taken at a level just above the motor decussation. They show that the predominant type of axis-cylinder is one of small caliber. Magnification 720X.


MAN
DOG
CAT
COW
RABBIT
RAT
MOUSE
OPOSSUM

group. Although the rabbit is a larger animal than the rat, the average diameter of its pyramidal tract fibers is less. There is no notable difference between the bundles of the rat and oppossum in spite of the fact that the latter is an arboreal animal and uses its digits for grasping and swinging in the trees.

In considering the placement of the fibers, we found no accumulation of either small or large fibers in one particular part of the pyramidal tract in the group examined. These fibers are equally dispersed and mixed throughout. It is stated that in man the fibers which go to the leg occupy the lateral and those which go to the arm and face the medial part of the pyramidal tract. If this be true, then the fibers which extend to the lower extremities are of the same diameters as those which supply the upper muscles of the body.

Because the fibers of the pyramidal tract are so characteristically small and many are unmyelinated, the application of the Marchi stain, although widely used in studying the pyramidal system, will not be highly efficient, especially on the lower mammals. As a case in point, one can compare the investigation of King ('10) on the rat with our findings in the same animal. She removed the motor cortex in the rat and allowed 10 to 14 days for degeneration to occur. The Marchi stain was then applied to various parts of the central nervous system. At a level directly above the decussation, she counted 901 degenerated fibers and concluded from this that the primary motor path was unimportant. In contrast to this, we counted 73,000 or over eighty times more normal fibers at an identical level.

Since more and more evidence is accumulating that the size of nerve fibers is related to the speed of conduction, it would seem safe to say that the pyramidal tract is on the whole a bundle which conducts slowly because of the great preponderance of minute fibers.

Having discussed the number and size of pyramidal tract axis-cylinders, the question as to their cells of origin can next be considered.

Cells of origin of pyramidal tract fibers

The discussion on the cells of origin of the pyramidal tract fibers is based on the belief that the size of the fibers should be proportional to the size of their cell bodies.

Some investigators advocate that the pyramidal tract is made up of fibers concerned with the control of the finer digital movements (Barron, '34; Foerster, '31 and '36; Kennard, '39; Leyton and Sherrington, '17; Tower and Hines, '35). This function reaches its highest development in primates and is described as being controlled by Betz cells located in area 4 (Brodmann). These cells are large, have unusual excitability and would be expected to give origin to myelinated fibers of large size. The only distinguishing feature which the pyramidal tract of man possesses from other animals of the series is in a certain number of fibers of large diameter (over 10 μ) although the dog has a few thick fibers of this dimension. The number of these large fibers agrees more closely with the number of Betz cells. On the basis of this reasoning, it would seem logical to conclude that the Betz cells and the largest pyramidal tract fibers of man may be specifically concerned with the control of the finer digital movements. Häggquist's investigation on the monkey ('37) would lend support to this view. After removal of various parts of the right gigantopyramidalis area in four monkeys, he found that the largest fibers disappeared when sufficient time was allowed for degeneration. He concluded that five-sixths of the fibers present in the pyramid of the medulla are not cortico-spinal but belong to other systems. The origin of the remainder of the smaller fibers which form such a large part of the pyramidal tract is speculative. Häggquist's work suggests a region or regions outside of area 4. Most evidence points to area 6 (Bucy, '33 and '34; Foerster, '36; Kennard, '35). Whatever part of the cortex may be concerned, it is believed by us that the cells of origin should be small in size to correspond with the caliber of the predominant fiber.

It is more difficult to discuss the cells of origin of the pyramidal tract in the mammals ranking below man in the series. The results of stimulation experiments on the motor region are not in accord. These animals all possess motor areas but they are better developed in the cat and dog than in the rabbit, rat, mouse, and opossum. Cells resembling those found in the motor cortex of higher mammals are probably present in all. Whether these give sole origin to all the fibers of the pyramidal tract is a phase of research that has not been attempted as far as we have been able to ascertain. Von Bonin ('38) has reported that the size of the nuclei of the giant pyramidal cells are larger in man than in the cat and that the more complex the brain the greater the variability in the size of the cortical cells. Our findings on the pyramidal tract agree with this view as the fibers in the lower animals are small and more or less uniform in diameter. We feel that the dog should have some cells of origin of very large caliber but that the majority should be small to correspond with the predominant fiber. The cells of origin of the pyramidal tract of the cat, rat, opossum, and mouse should also be on the whole of small size to coincide with their axis-cylinders.

CONCLUSIONS

1. The characteristic axis-cylinder of the pyramidal tract in a series of eight mammals is one of small caliber.

2. Man possesses the greatest number of pyramidal tract fibers followed in order by the dog, cat, rabbit, opossum, rat, and mouse.

3. The distinguishing features of the pyramidal tract of man are the following: It has the greatest area, the largest number of fibers, and the thickest axis-cylinders.

4. The small animals possess a relatively large number of fibers when their mass is taken into consideration.

5. In the smaller animals, there is a greater number of pyramidal tract fibers per unit area.

6. The sizes of the axis-cylinders in the pyramidal tract are not proportional to the mass of the animal.

7. Small, intermediate and large-sized fibers are equally dispersed and mixed throughout the pyramidal tract.

8. The vast majority of the cells of origin of the pyramidal tract fibers are probably small in size.

9. The conduction rate over the pyramidal tract should be on the whole slow.

10. The Marchi technic should not be reliable on the pyramidal tract of lower mammals.

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